

Data, loss and gradient descent

1 A model adds up evidence

Each clue gets a weight. Add the weights, squash the total into 0–1 with the S-curve. That's logistic regression.



2 Score probabilities with log loss

Penalty = surprise. It rewards honest confidence and punishes confident mistakes very hard.



3 Roll downhill

Gradient descent: feel the slope of the loss, take a small step down, repeat. Step size matters.

4 It works

Twenty lines of NumPy get within 1% of the best possible log loss on new alerts.



5 Weights are readable

Learned odds multipliers: +0.1 threat-intel score ×1.85; approved IT tooling ×0.11.

6 Test on data it never saw

Flexible models can memorise. Only held-out data tells learning from memorising.

